Employee Attrition Prediction Report

Executive Summary:

This project focus on predicting employee attrition using split, train and testing the given data set. The goal is to generate a consolidated report on attrition driving factors, so that HR can take action.
3 classifications models are used. The best performance model is selected in the end.

Dataset Overview:

Dataset: IBM HR Analytics Employee Attrition dataset
Records: 1470
Target Variable: Attrition
Feaures: Job Role, Department, Mothly income, Age, Work-life balance, job satisfaction, etc.
Data Preprocessing: Removed irrelevant columns. Removed AttritionColumn from X and assigned it to Y. Handled missing values. Encoding and standard scaling done. Train test split is done.

Key Insights:

	Age	Attrition	BusinessTravel	DailyRate	Department
0	41	Yes	Travel_Rarely	1102	Sales
1	49	No	Travel_Frequently	279	Research & Development
2	37	Yes	Travel_Rarely	1373	Research & Development
3	33	No	Travel_Frequently	1392	Research & Development
4	27	No	Travel_Rarely	591	Research & Development

	DistanceFromHome	Education	EducationField	EmployeeCount	EmployeeNumber
0	1	2	Life Sciences	1	1
1	8	1	Life Sciences	1	2
2	2	2	Other	1	4
3	3	4	Life Sciences	1	5
4	2	1	Medical	1	7

	...	RelationshipSatisfaction	StandardHours	StockOptionLevel
0	...	1	80	0
1	...	4	80	1
2	...	2	80	0
3	...	3	80	0
4	...	4	80	1

	TotalWorkingYears	TrainingTimesLastYear	WorkLifeBalance	YearsAtCompany
0	8	0	1	6

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1           10              3           3          10
2            7              3           3           0
3            8              3           3           8
4            6              3           3           2

  YearsInCurrentRole  YearsSinceLastPromotion  YearsWithCurrManager
0                  4                        0                     5
1                  7                        1                     7
2                  0                        0                     0
3                  7                        3                     0
4                  2                        2                     2

[5 rows x 35 columns]
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1470 entries, 0 to 1469
Data columns (total 35 columns):
 #   Column                    Non-Null Count  Dtype 
---  ------                    --------------  ----- 
 0   Age                       1470 non-null   int64 
 1   Attrition                 1470 non-null   object
 2   BusinessTravel            1470 non-null   object
 3   DailyRate                 1470 non-null   int64 
 4   Department                1470 non-null   object
 5   DistanceFromHome          1470 non-null   int64 
 6   Education                 1470 non-null   int64 
 7   EducationField            1470 non-null   object
 8   EmployeeCount             1470 non-null   int64 
 9   EmployeeNumber            1470 non-null   int64 
 10  EnvironmentSatisfaction   1470 non-null   int64 
 11  Gender                    1470 non-null   object
 12  HourlyRate                1470 non-null   int64 
 13  JobInvolvement            1470 non-null   int64 
 14  JobLevel                  1470 non-null   int64 
 15  JobRole                   1470 non-null   object
 16  JobSatisfaction           1470 non-null   int64 
 17  MaritalStatus             1470 non-null   object
 18  MonthlyIncome             1470 non-null   int64 
 19  MonthlyRate               1470 non-null   int64 
 20  NumCompaniesWorked        1470 non-null   int64 
 21  Over18                    1470 non-null   object
 22  OverTime                  1470 non-null   object
 23  PercentSalaryHike         1470 non-null   int64 
 24  PerformanceRating         1470 non-null   int64 
 25  RelationshipSatisfaction  1470 non-null   int64 
 26  StandardHours             1470 non-null   int64 
 27  StockOptionLevel          1470 non-null   int64 
 28  TotalWorkingYears         1470 non-null   int64 
 29  TrainingTimesLastYear     1470 non-null   int64 
 30  WorkLifeBalance           1470 non-null   int64 
 31  YearsAtCompany            1470 non-null   int64 
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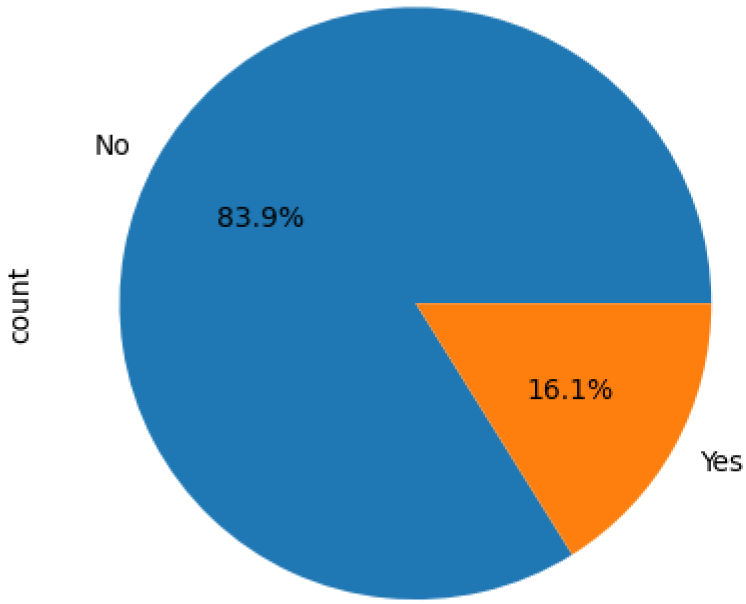
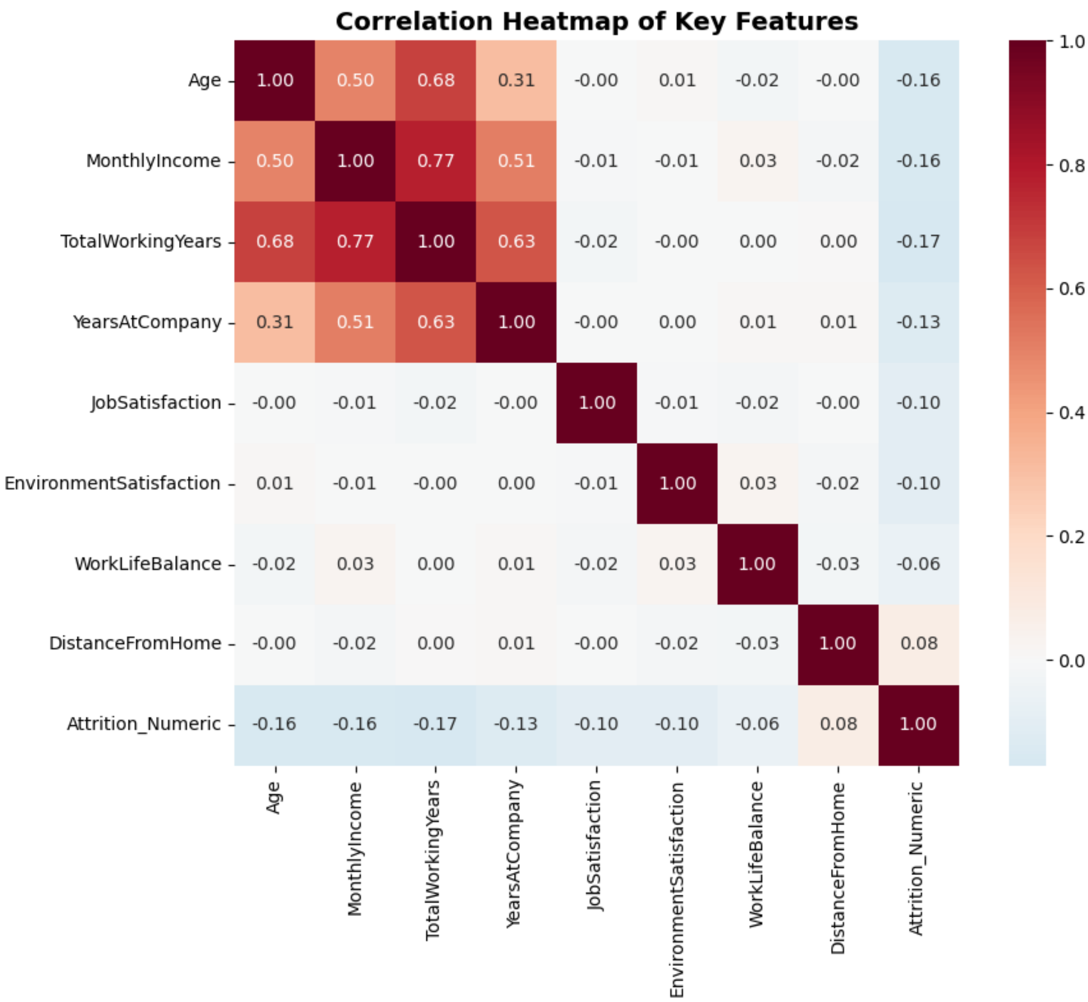
 32  YearsInCurrentRole        1470 non-null   int64
 33  YearsSinceLastPromotion   1470 non-null   int64
 34  YearsWithCurrManager      1470 non-null   int64
dtypes: int64(26), object(9)
memory usage: 402.1+ KB
None
Missing Values:
 Age                         0
Attrition                   0
BusinessTravel              0
DailyRate                   0
Department                  0
DistanceFromHome            0
Education                   0
EducationField              0
EmployeeCount               0
EmployeeNumber              0
EnvironmentSatisfaction     0
Gender                      0
HourlyRate                  0
JobInvolvement              0
JobLevel                    0
JobRole                     0
JobSatisfaction             0
MaritalStatus               0
MonthlyIncome               0
MonthlyRate                 0
NumCompaniesWorked          0
Over18                      0
OverTime                    0
PercentSalaryHike           0
PerformanceRating           0
RelationshipSatisfaction    0
StandardHours               0
StockOptionLevel            0
TotalWorkingYears           0
TrainingTimesLastYear       0
WorkLifeBalance             0
YearsAtCompany              0
YearsInCurrentRole          0
YearsSinceLastPromotion     0
YearsWithCurrManager        0
dtype: int64
Attrition
No     1233
Yes     237
Name: count, dtype: int64
```


No
83.9%
count
16.1%
Yes


Correlation Heatmap of Key Features
Age
MonthlyIncome
TotalWorkingYears
YearsAtCompany
JobSatisfaction
EnvironmentSatisfaction
WorkLifeBalance
DistanceFromHome
Attrition_Numeric
1.00 0.50 0.68 0.31 -0.00 0.01 -0.02 -0.00 -0.16
0.50 1.00 0.77 0.51 -0.01 -0.01 0.03 -0.02 -0.16
0.68 0.77 1.00 0.63 -0.02 -0.00 0.00 0.00 -0.17
0.31 0.51 0.63 1.00 -0.00 0.00 0.01 0.01 -0.13
-0.00 -0.01 -0.02 -0.00 1.00 -0.01 -0.02 -0.00 -0.10
0.01 -0.01 -0.00 0.00 -0.01 1.00 0.03 -0.02 -0.10
-0.02 0.03 0.00 0.01 -0.02 0.03 1.00 -0.03 -0.06
-0.00 -0.02 0.00 0.01 -0.00 -0.02 -0.03 1.00 0.08
-0.16 -0.16 -0.17 -0.13 -0.10 -0.10 -0.06 0.08 1.00
1.0
0.8
0.6
0.4
0.2
0.0

```
? HR RISK DISTRIBUTION:
Risk_Level
Low Risk      241
Medium Risk    36
High Risk      17
Name: count, dtype: int64

? Attrition Rate by Risk Level:
Risk_Level
Low Risk      0.087137
Medium Risk   0.361111
High Risk     0.764706
Name: Actual, dtype: float64

? TOP 5 HIGH RISK EMPLOYEES:
    Attrition_Prob Risk_Level
357       0.923953  High Risk
911       0.888170  High Risk
688       0.885017  High Risk
762       0.848979  High Risk
301       0.764567  High Risk

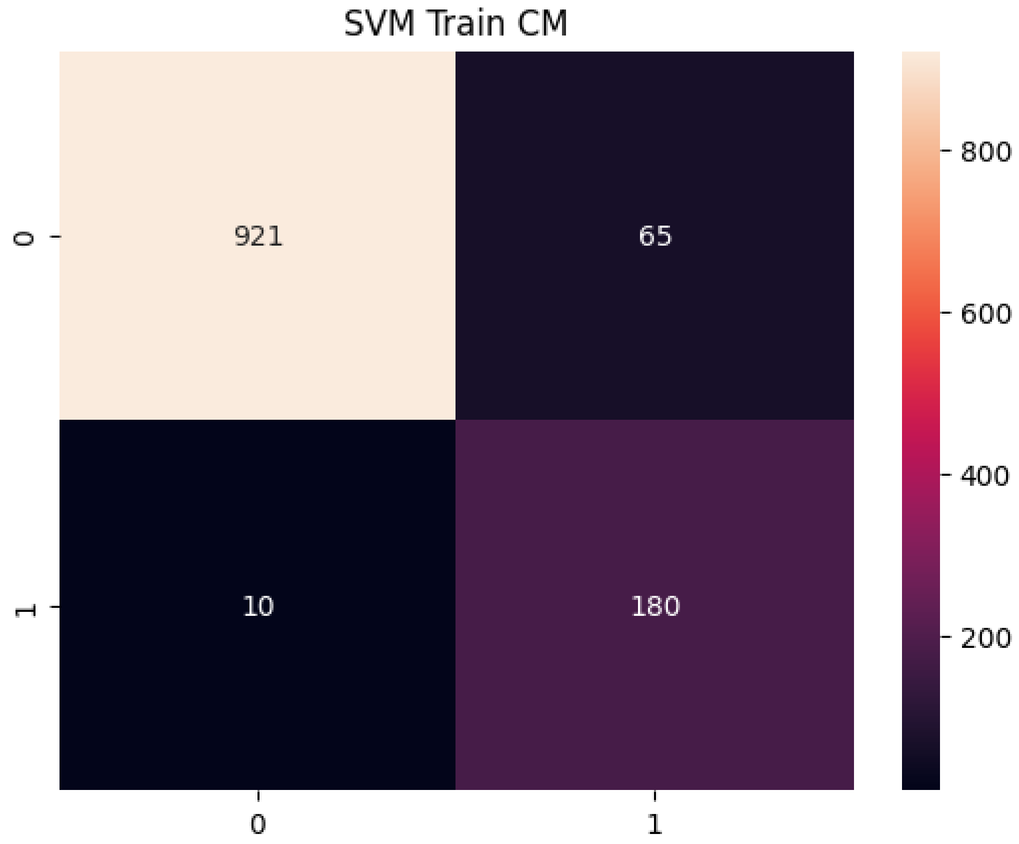
? Key Attrition Drivers:

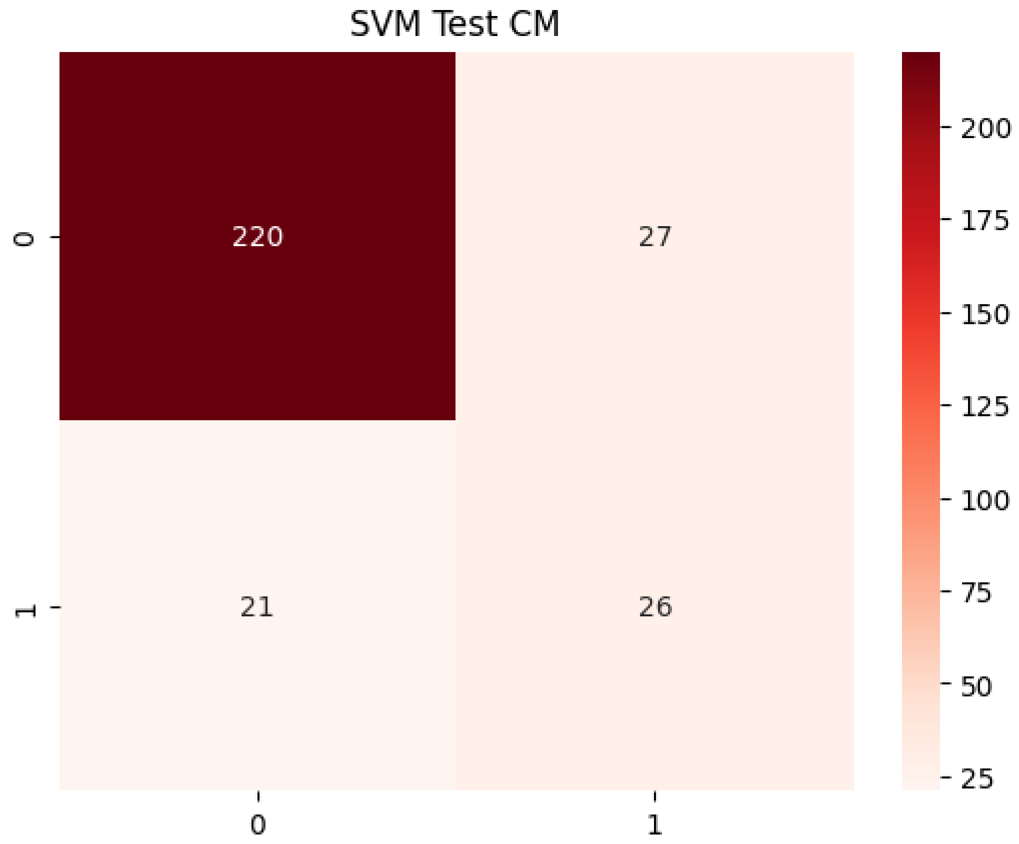
Top Risk Factors:
DistanceFromHome    0.040668
HourlyRate          0.044646
Predicted           0.446942
Attrition_Prob      0.510422
Actual              1.000000
Name: Actual, dtype: float64

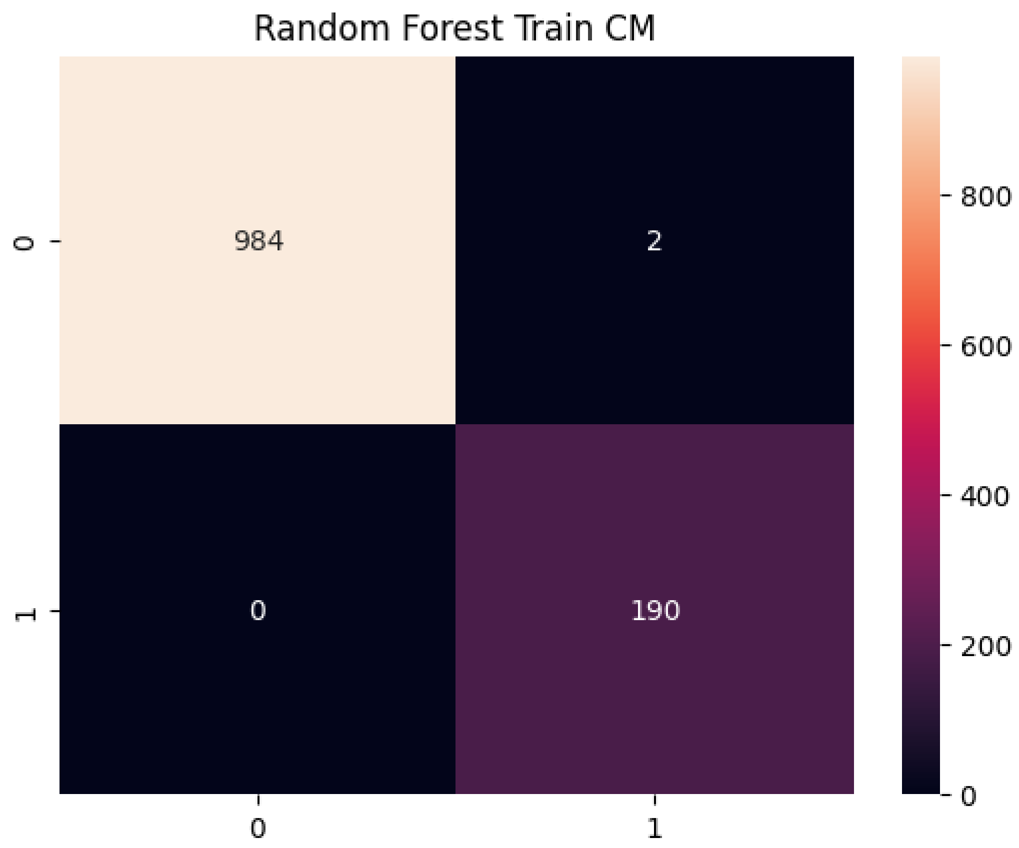
Protective Factors:
JobLevel               -0.180605
MonthlyIncome          -0.158062
JobInvolvement         -0.149344
YearsInCurrentRole     -0.136652
TrainingTimesLastYear  -0.131820
Name: Actual, dtype: float64

? Best Model: SVM
Model saved as model.pkl
? In Pickle file Best Model Name: SVM
               Model  Accuracy  Precision    Recall        F1   ROC-AUC  \
0                SVM  0.836735   0.490566  0.553191  0.520000  0.816005
1      Random Forest  0.806122   0.421875  0.574468  0.486486  0.794039
2  Gradient Boosting  0.829932   0.465116  0.425532  0.444444  0.801016

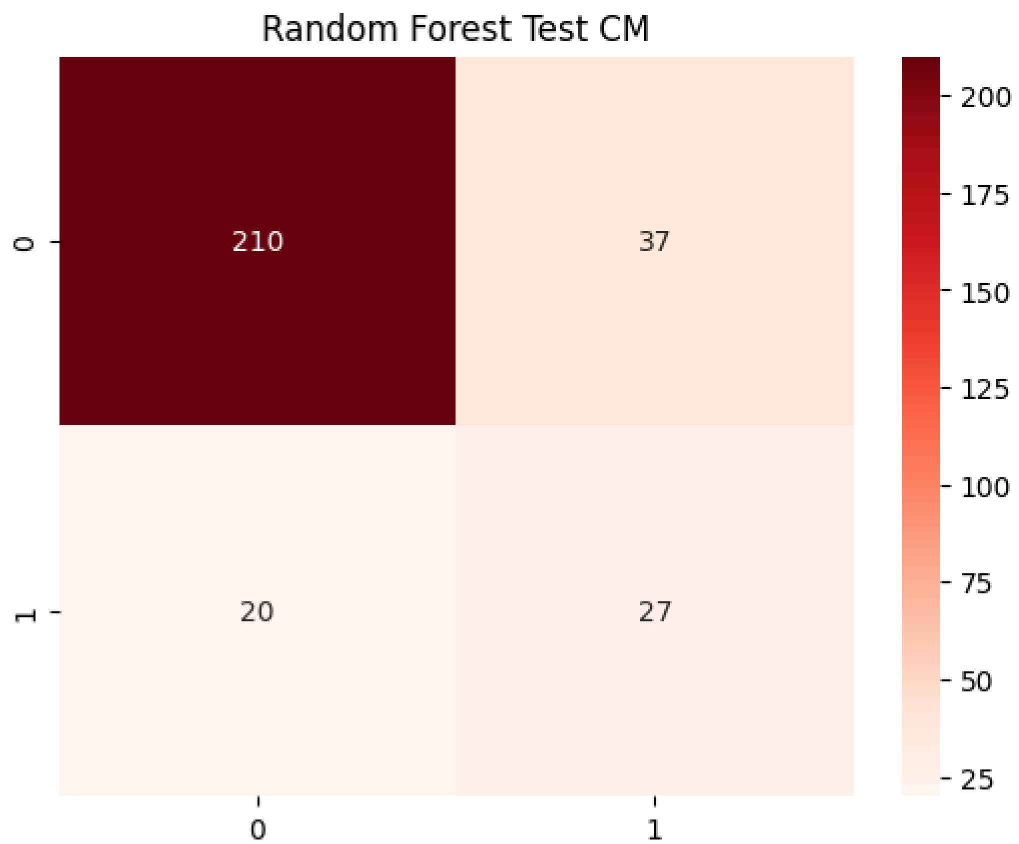
     PR-AUC  confusion_matrix_train  confusion_matrix_test
0  0.567844  [[921, 65], [10, 180]]  [[220, 27], [21, 26]]
1  0.454538    [[984, 2], [0, 190]]  [[210, 37], [20, 27]]
2  0.456457    [[985, 1], [91, 99]]  [[224, 23], [27, 20]]
```


SVM Train CM
921
65
10
180
0
1
0
1
800
600
400
200


SVM Test CM
220
27
21
26
0
1
0
1
200
175
150
125
100
75
50
25

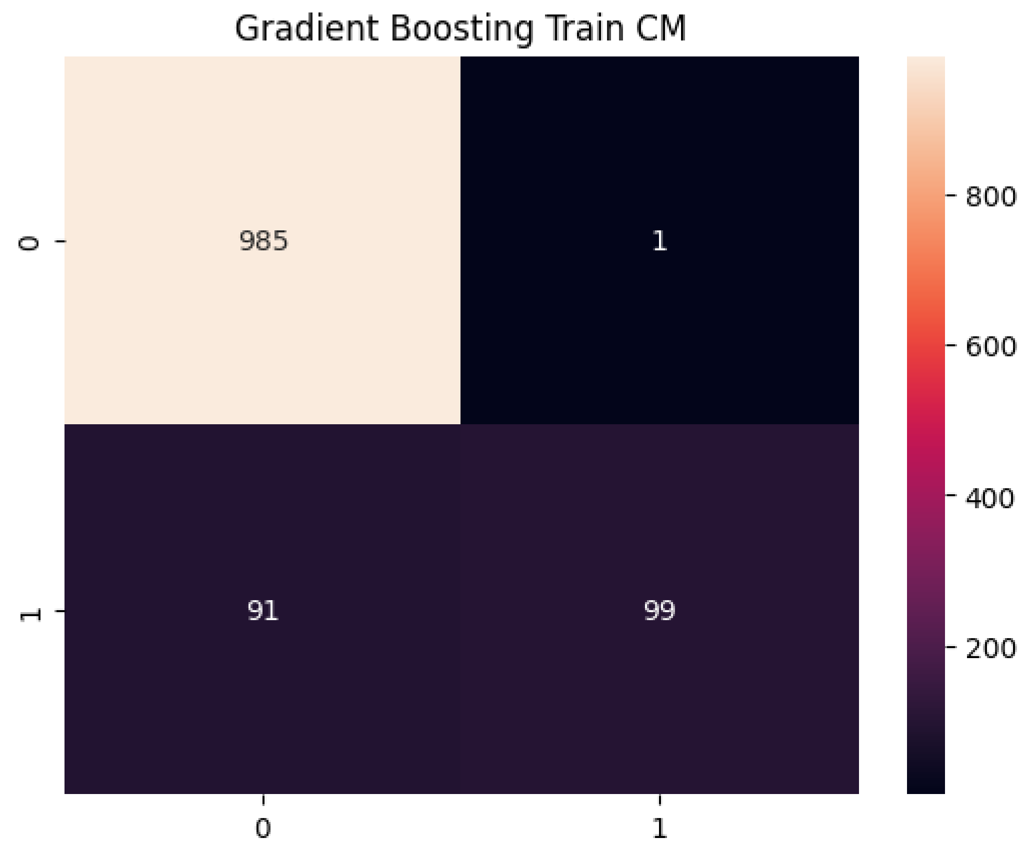


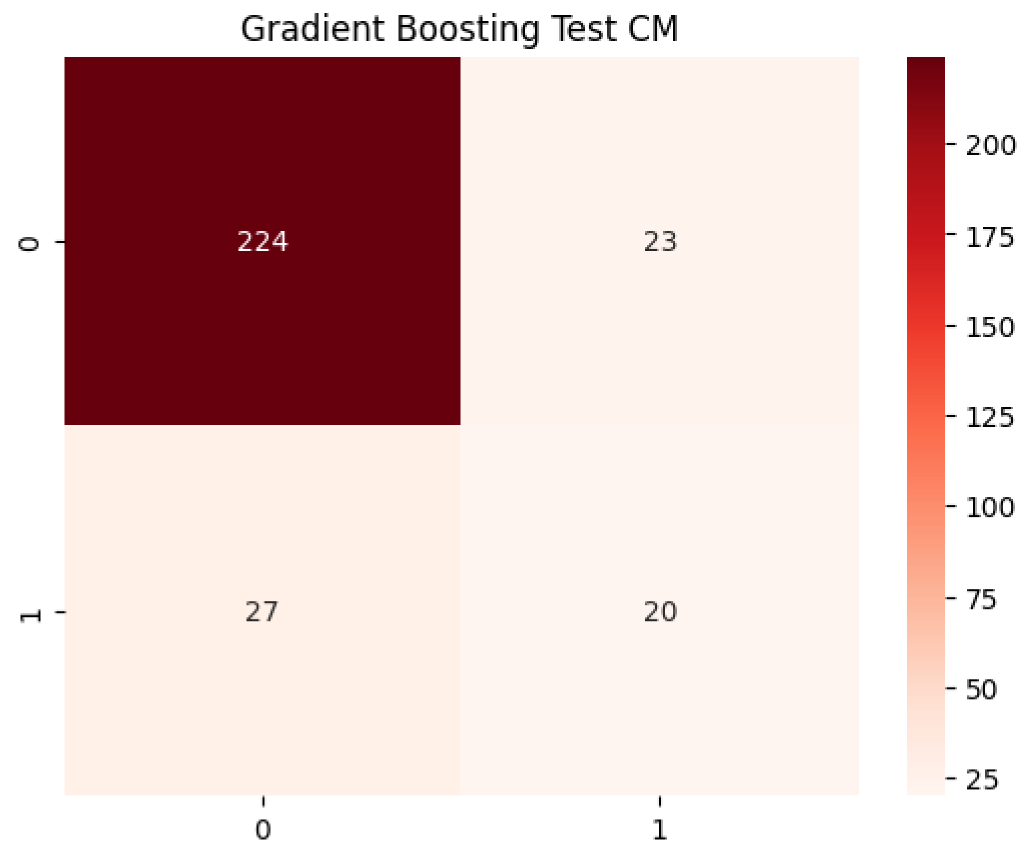
Random Forest Train CM
984
2
0
190
0
1
0
1
800
600
400
200
0



Random Forest Test CM
210
37
20
27
0
1
0
1
200
175
150
125
100
75
50
25

Gradient Boosting Train CM



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Cleaned data saved to data/cleaned_employee_attrition.csv
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